



■ CS181: Artificial Intelligence

Lecture 15: Transformer & BERT

Sibei Yang





Outline

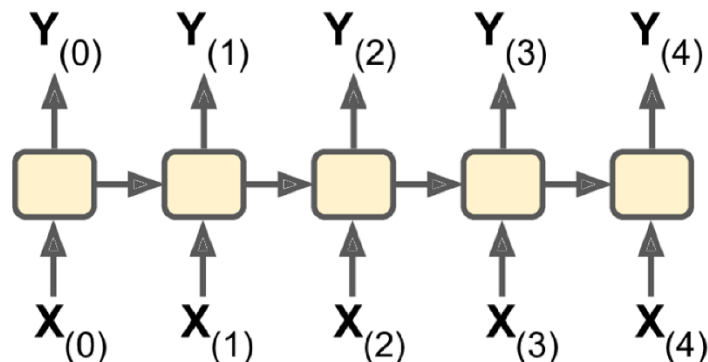


- Transformer
 - Self-attention
 - Multi-head Attention
 - Transformer Layer
 - Transformer Encoder & Decoder
- BERT
 - Transformer?
 - Pre-training
 - Fine-tuning



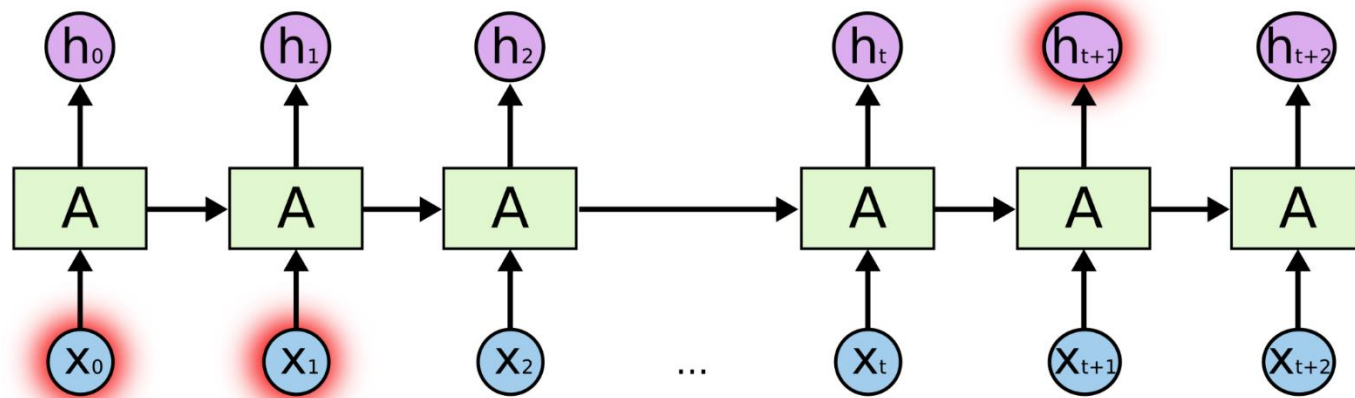
Sequence Modeling

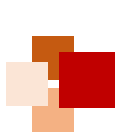
- RNN, LSTM, GRU



- Parallel
- Learning long-term dependencies is hard

Attention



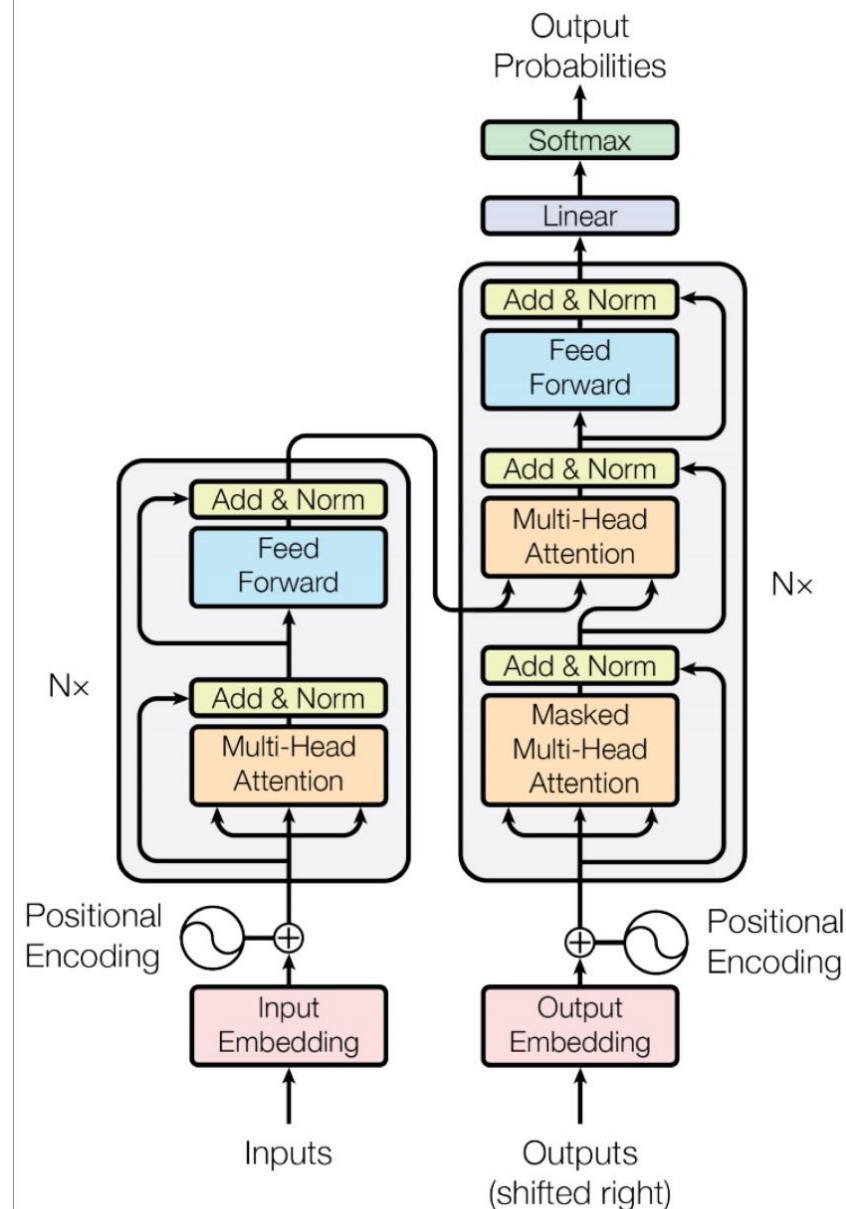


Transformer



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- Transformer [Vaswani et al., 2017]
- Non-recurrent seq2seq model
 - Machine translation
 - Encoder
 - Decoder
- Encoder only



"Attention Is All You Need", Vaswani et al, 2017. <https://arxiv.org/pdf/1706.03762.pdf>

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Dot-Product Attention



- Input: a query q , a set of (key, value) (K, V)
- Output: weighted sum of the values, dependent on each query
 - The weight is computed via the dot-product.

$$A(q, K, V) = \sum_i \frac{e^{q \cdot k_i}}{\sum_j e^{q \cdot k_j}} v_i$$



■ Dot-Product Attention



- Queries: $[q_1, q_2, q_3, \dots] \longrightarrow Q$

$$A(q, K, V) = \sum_i \frac{e^{q \cdot k_i}}{\sum_j e^{q \cdot k_j}} v_i$$



$$A(Q, K, V) = \text{softmax}(QK^T)V$$

Softmax over each row

$$[|Q|, d_k] \times [d_k, |K|] \times [|K|, d_v] \rightarrow [|Q|, d_v]$$



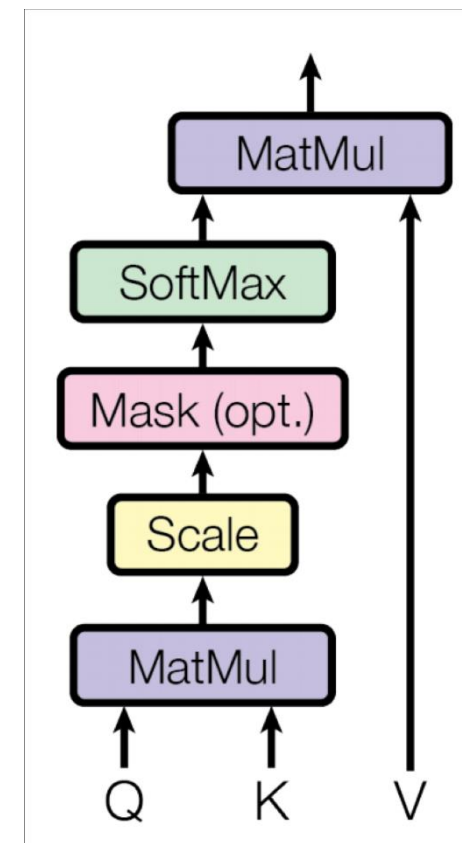
■ Dot-Product Attention

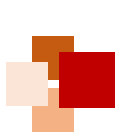


- When the input is large, the softmax function may have an extremely small gradient, hard for efficient learning

- Scaling:

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$



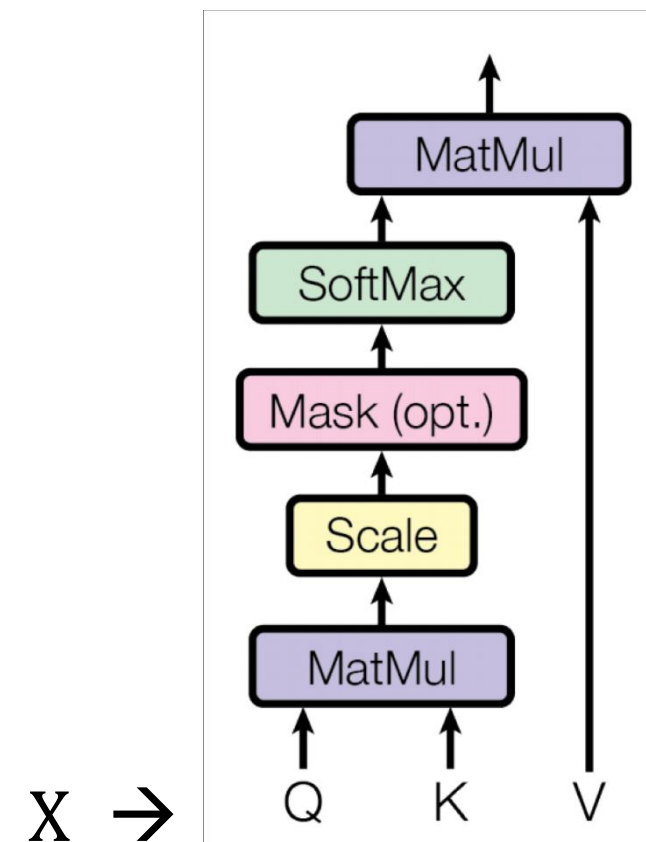


Self-attention



- An attention mechanism relating different positions of a single sequence in order to compute a representation of the same sequence.
- $X \rightarrow Q, K, V$

$$A(Q, K, V) = \text{softmax}\left(\frac{QK^T}{\sqrt{d_k}}\right)V$$

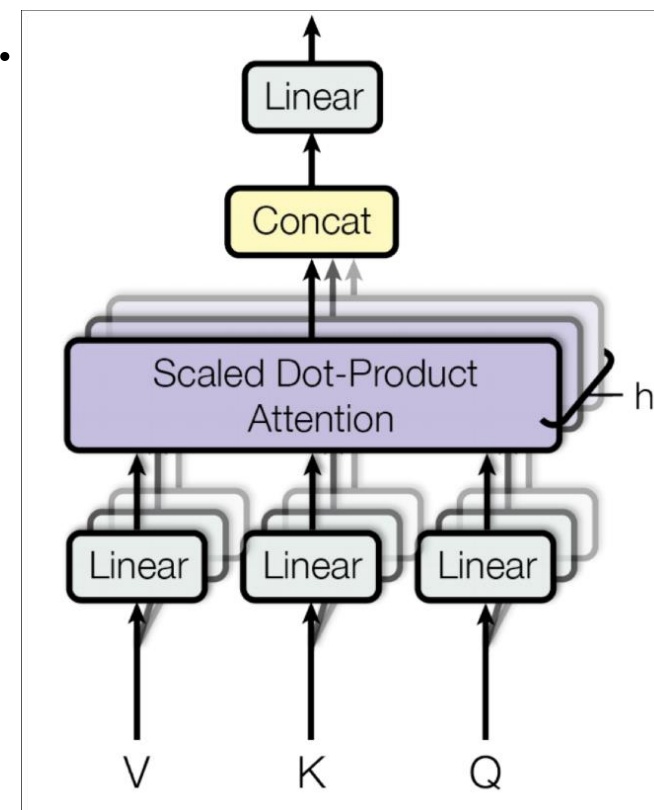


Multi-head self-attention

- A single attention head
- Multi-head attention
 - allows the model to jointly attend to information from different representation subspaces at different positions.

$$\text{MultiHead}(Q, K, V) = \text{Concat}(\text{head}_1, \dots, \text{head}_h)W^O$$

$$\text{where: } \text{head}_i = \text{Attention}(QW_i^Q, KW_i^K, VW_i^V)$$



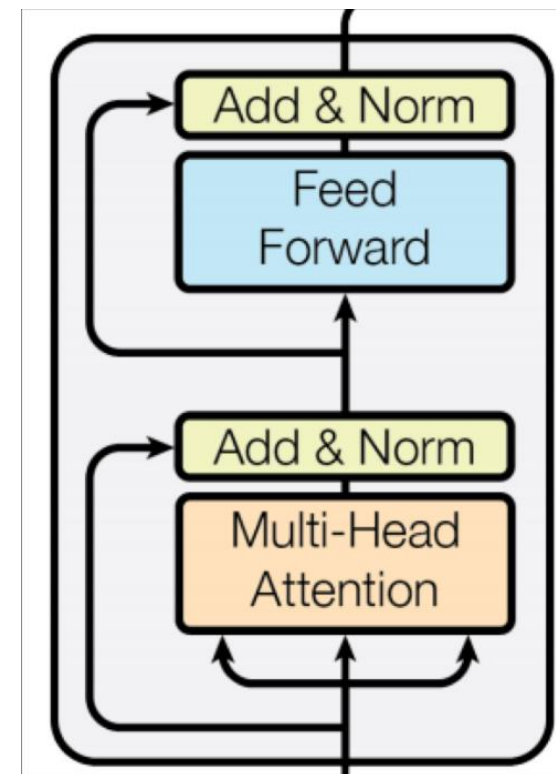
Transformer Layer



- Each Transformer layer consists of two sub-layers
 - Multi-head self-attention layer
 - Fully connected feed-forward network
- Each sub-layer adopts a residual connection and a layer normalization

LayerNorm($x + \text{Sublayer}(x)$)

$$\text{output} = \frac{x - \mu}{\sqrt{\sigma} + \epsilon} \times \gamma + \beta$$



Positional Encoding



- To preserve the position information, a sinusoid-wave-based positional encoding is applied and summed with the embedding output.

$$PE_{(pos,2i)} = \sin(pos/10000^{2i/d_{model}})$$

$$PE_{(pos,2i+1)} = \cos(pos/10000^{2i/d_{model}})$$

Others?

$$PE = [0, 1, 2, \dots, T-1]$$
$$PE = \left[\frac{0}{T}, \frac{1}{T}, \frac{2}{T}, \dots, \frac{T-1}{T} \right]$$

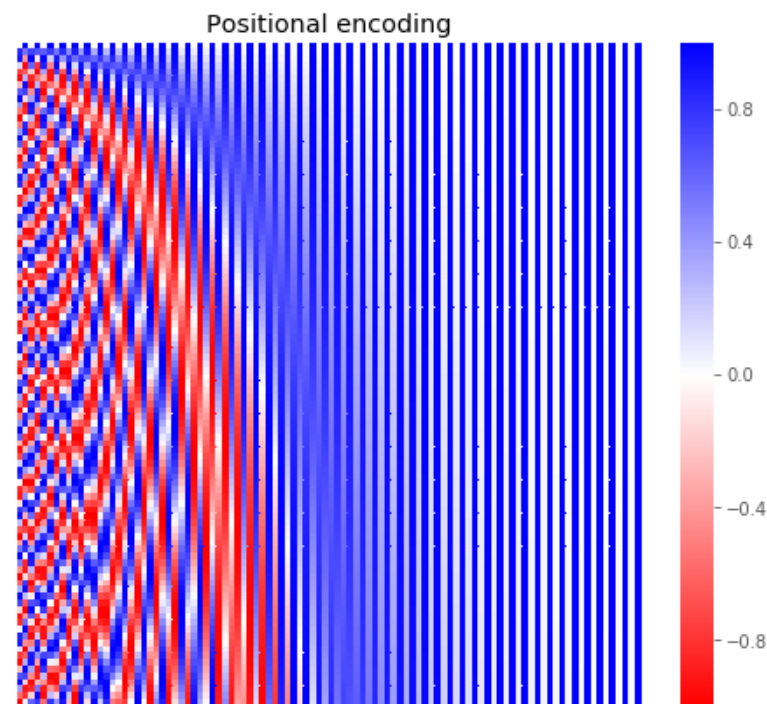


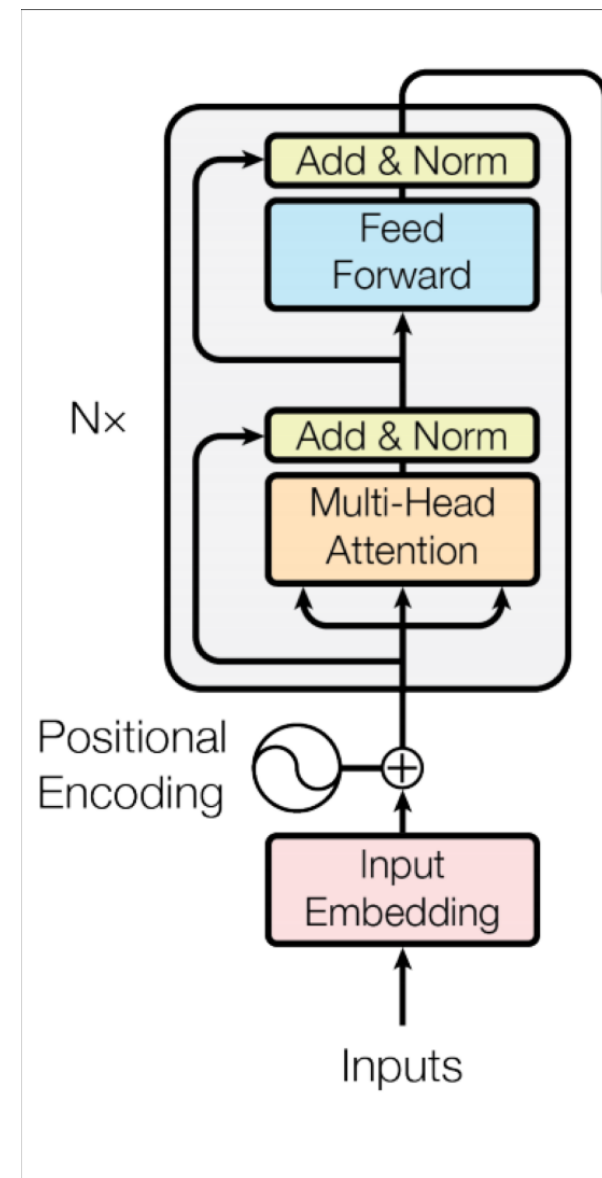
Figure from <http://www.batisengul.co.uk/post/2019-03-12-from-zero-to-state-of-the-art-nlp-part-two/>



Transformer Encoder



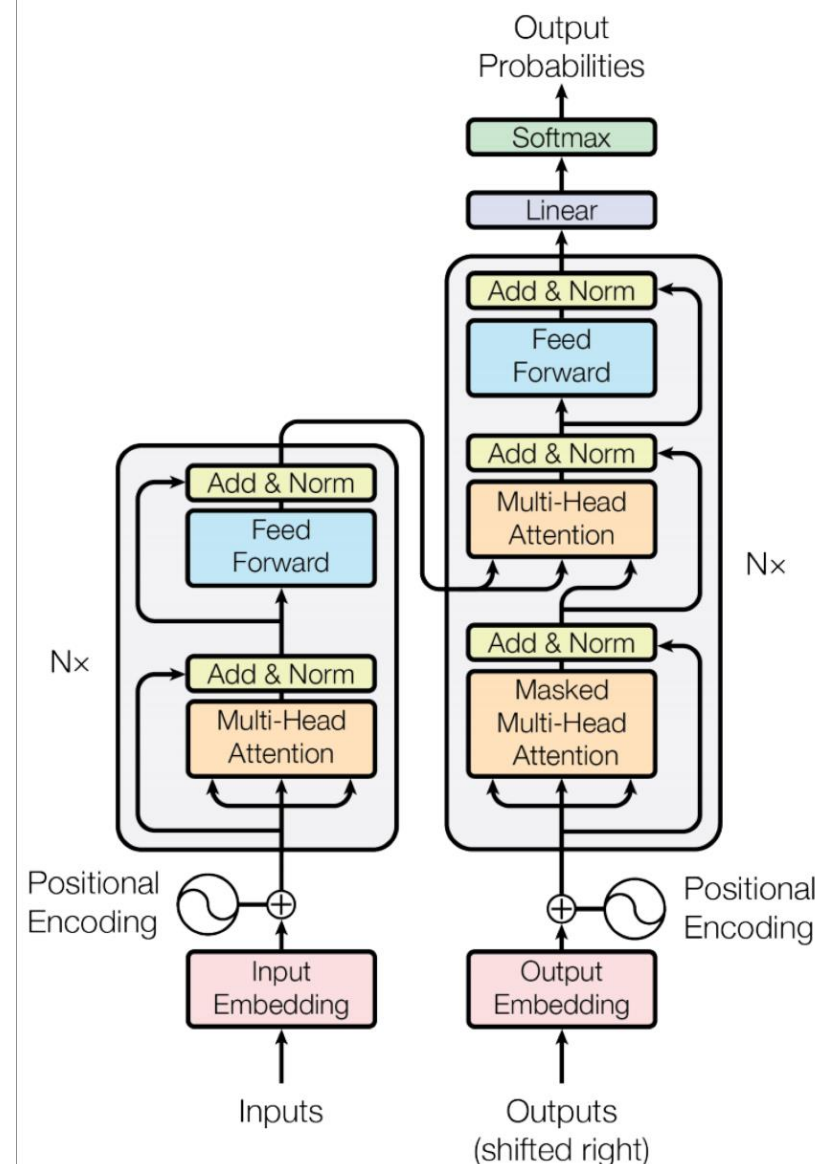
A stack of $N=6$
Transformer layers



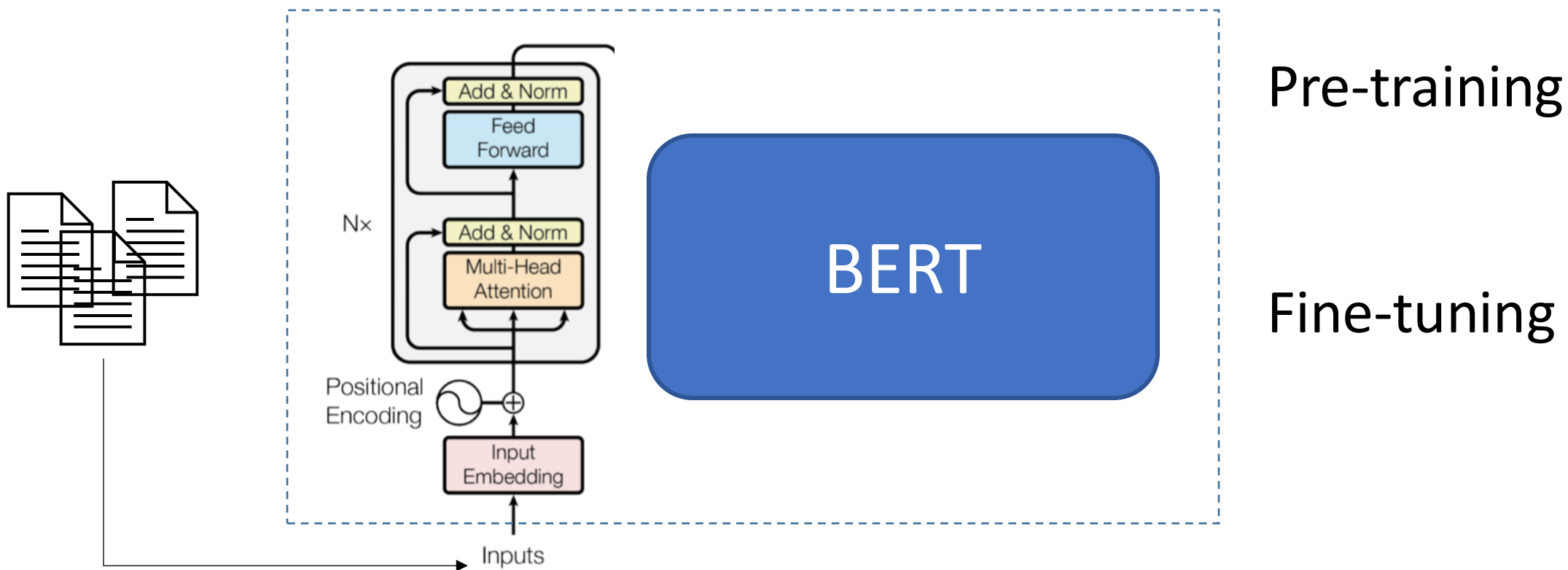
Transformer Decoder



- Mask: first multi-head attention sub-layer is modified to prevent positions from attending to subsequent positions.
- Cross-attention: Encoder $\rightarrow$ K & V
- A stack of $N=6$ Transformer layers



Bidirectional Encoder Representations from Transformers [Devlin, Jacob, et al. 2018]



Devlin, Jacob, et al. "Bert: Pre-training of deep bidirectional transformers for language understanding", 2018.

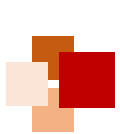


Contextual Representation



- Word embeddings
word2vec, GloVe
- Context free manner
'Bank'?
- Learn in the tasks
river-bank, money-bank...
- Train contextual representations on text corpus!!!





Contextual Representation



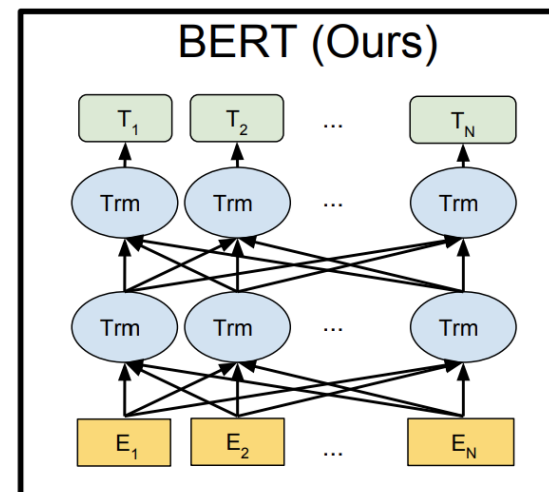
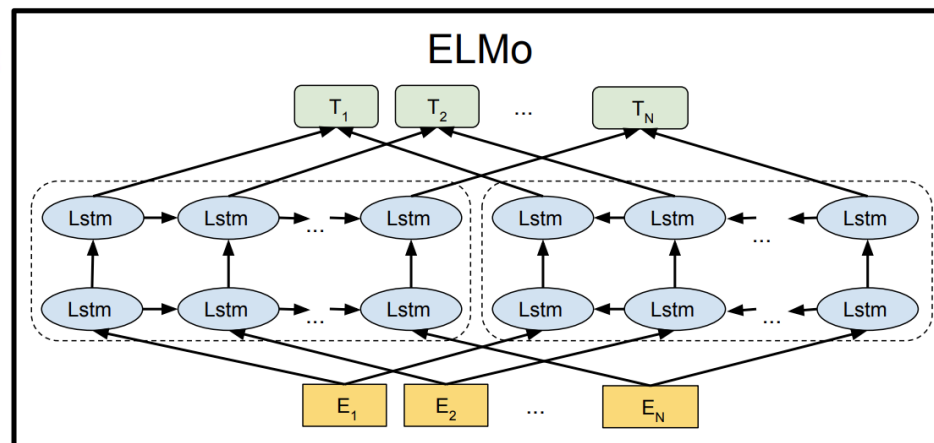
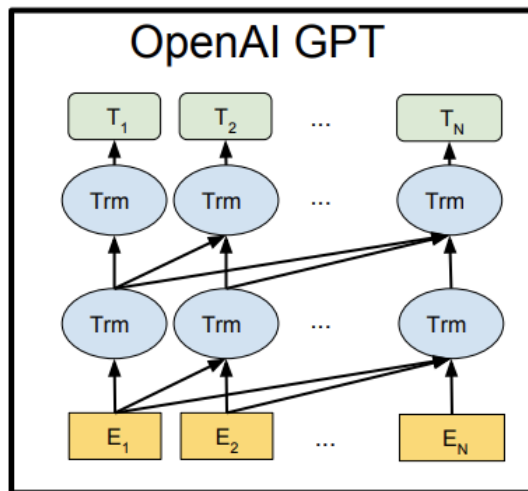
- Train contextual representations on text corpus

...

ELMo

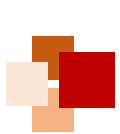


GPT



BERT





Embeddings from Language Model (ELMo)



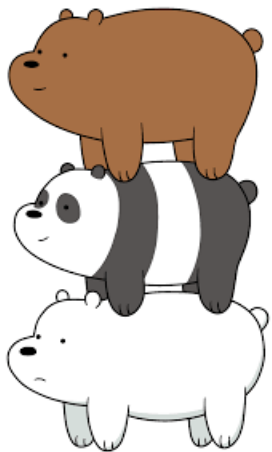
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- RNN-based language models

“I need a fan ...”



We bare bears



“...fan to cool the heat”

Hmm?

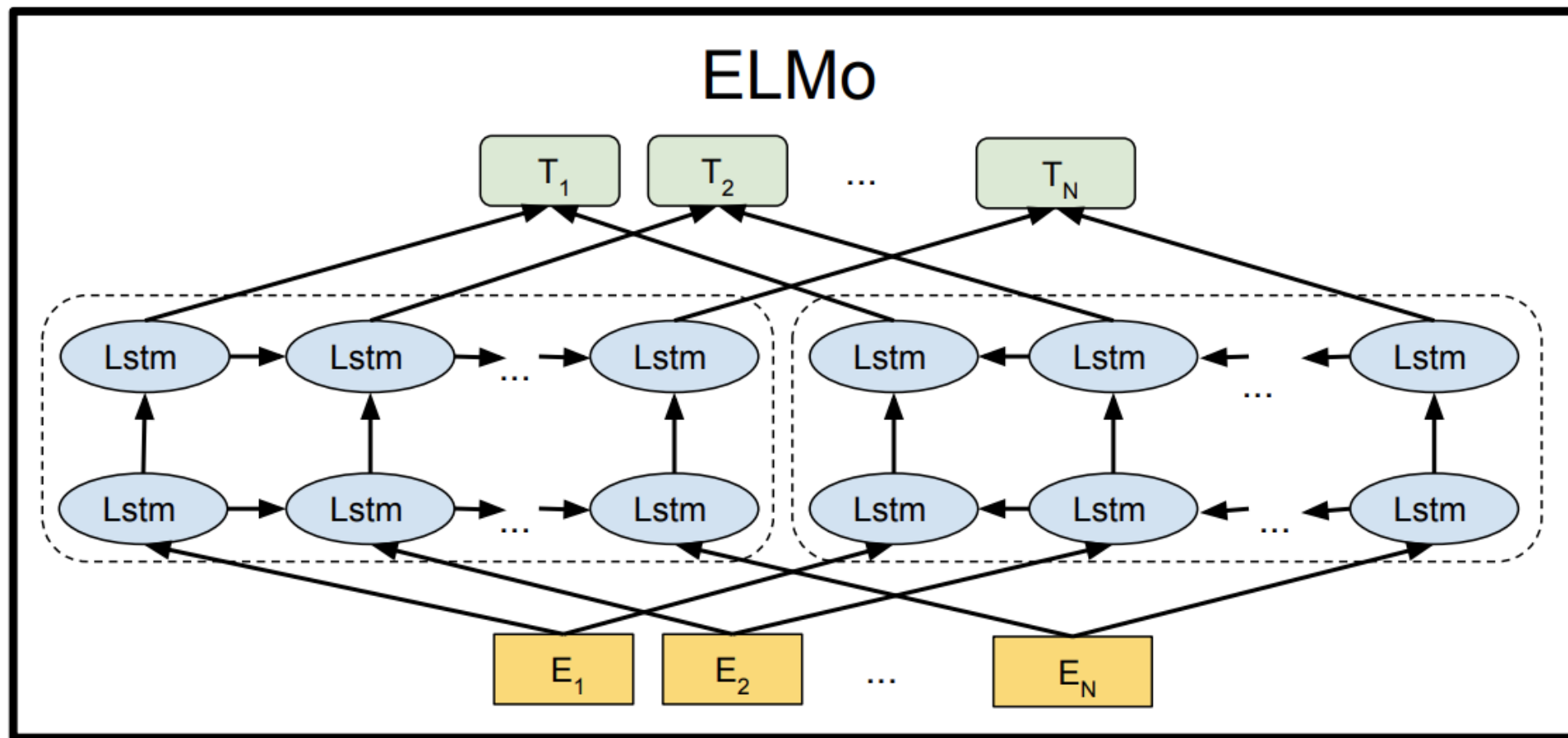




Embeddings from Language Model (ELMo)



- RNN-based bidirectional language models



“I need a fan to cool the heat”



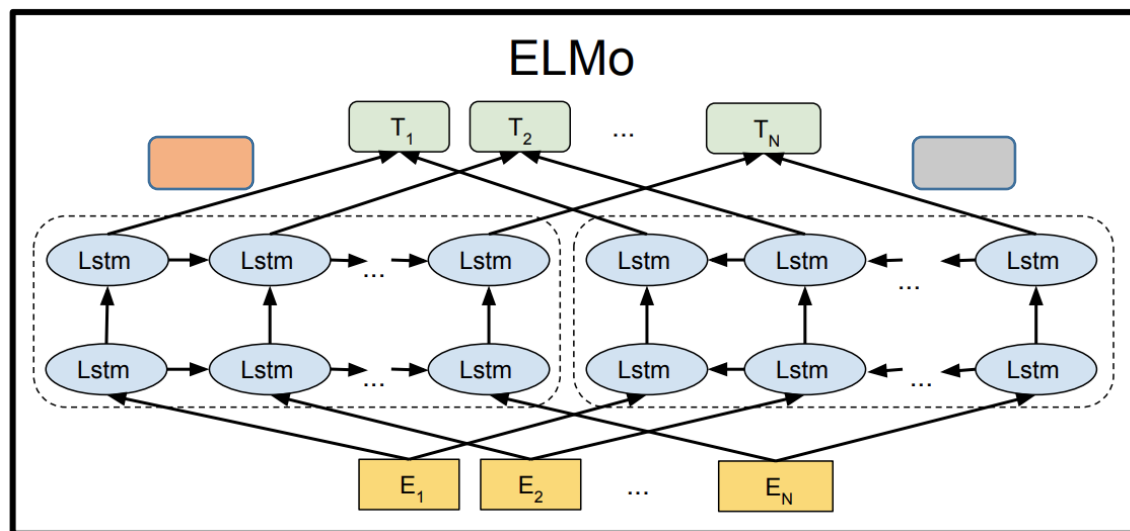


Embeddings from Language Model (ELMo)



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- RNN-based language models (trained from lots of sentences)



Pretraining

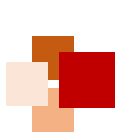
$$T_n = \alpha_1 L2R_n + \alpha_2 R2L_n$$

Learned with the down stream tasks

Task-specific combination of the representations



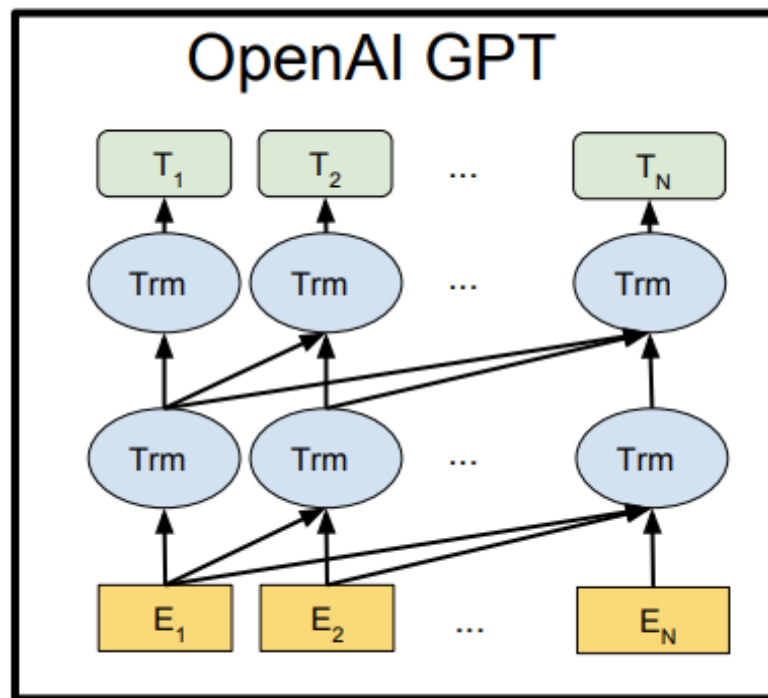
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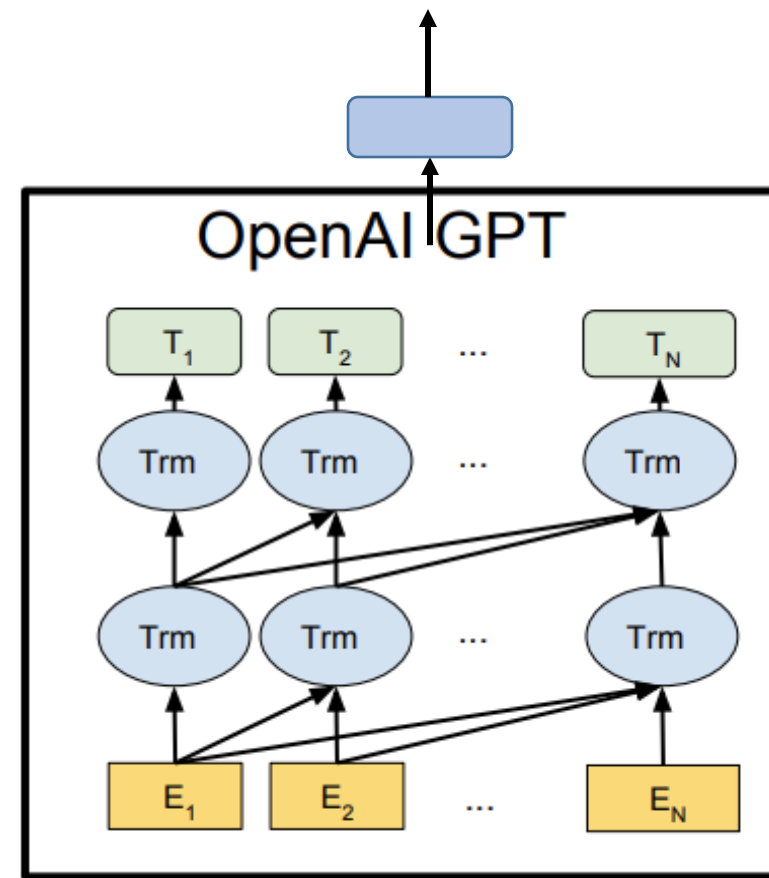
Generative Pre-Training (GPT)



A stack of $N=12$ Transformer layers (Decoder)



Pretraining



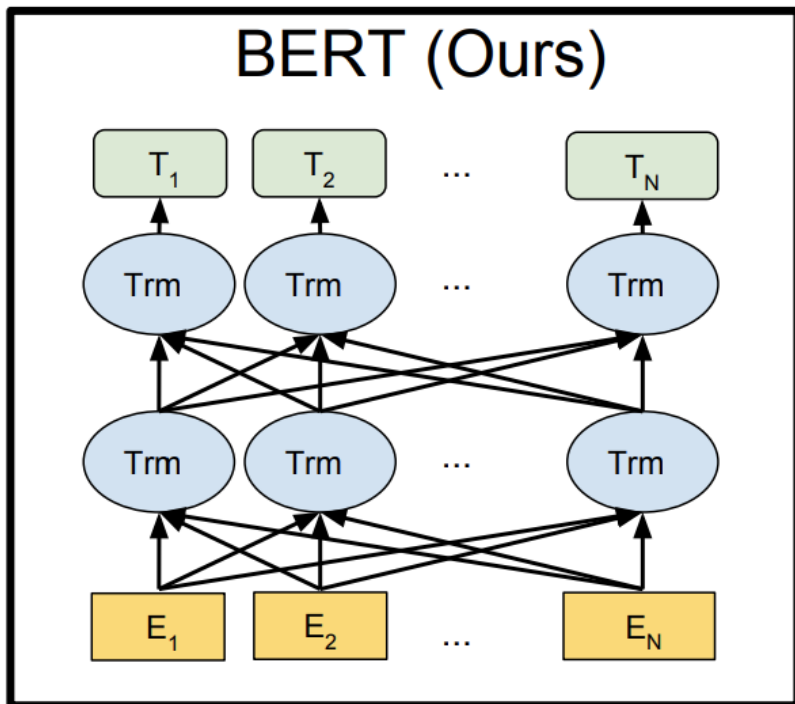
Finetuning



Bidirectional Encoder Representation from Transformer (BERT)



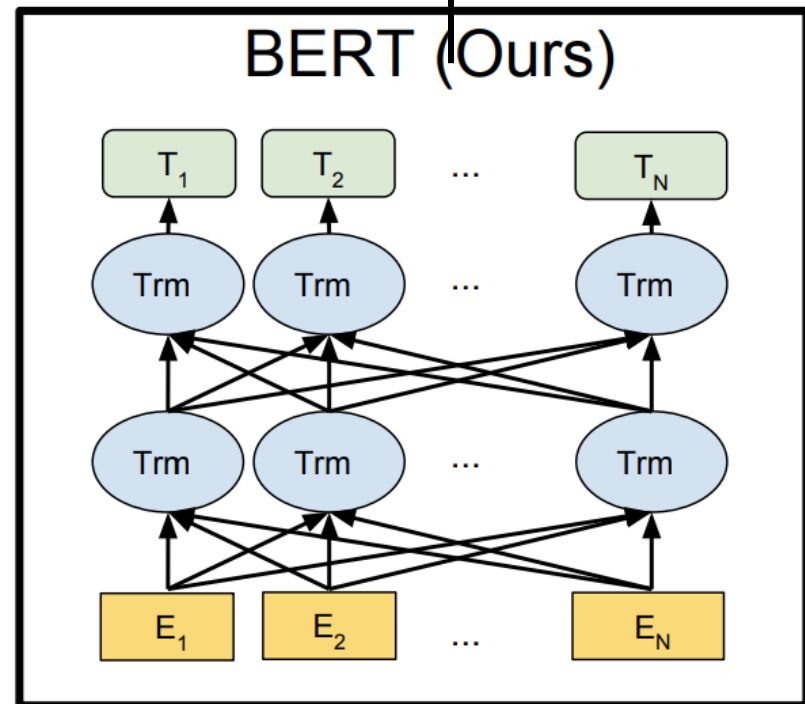
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Pretraining

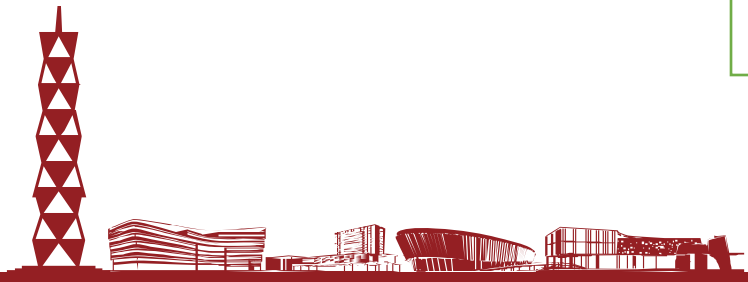
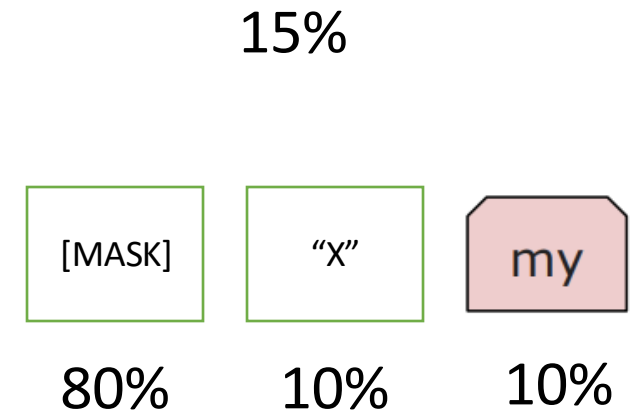
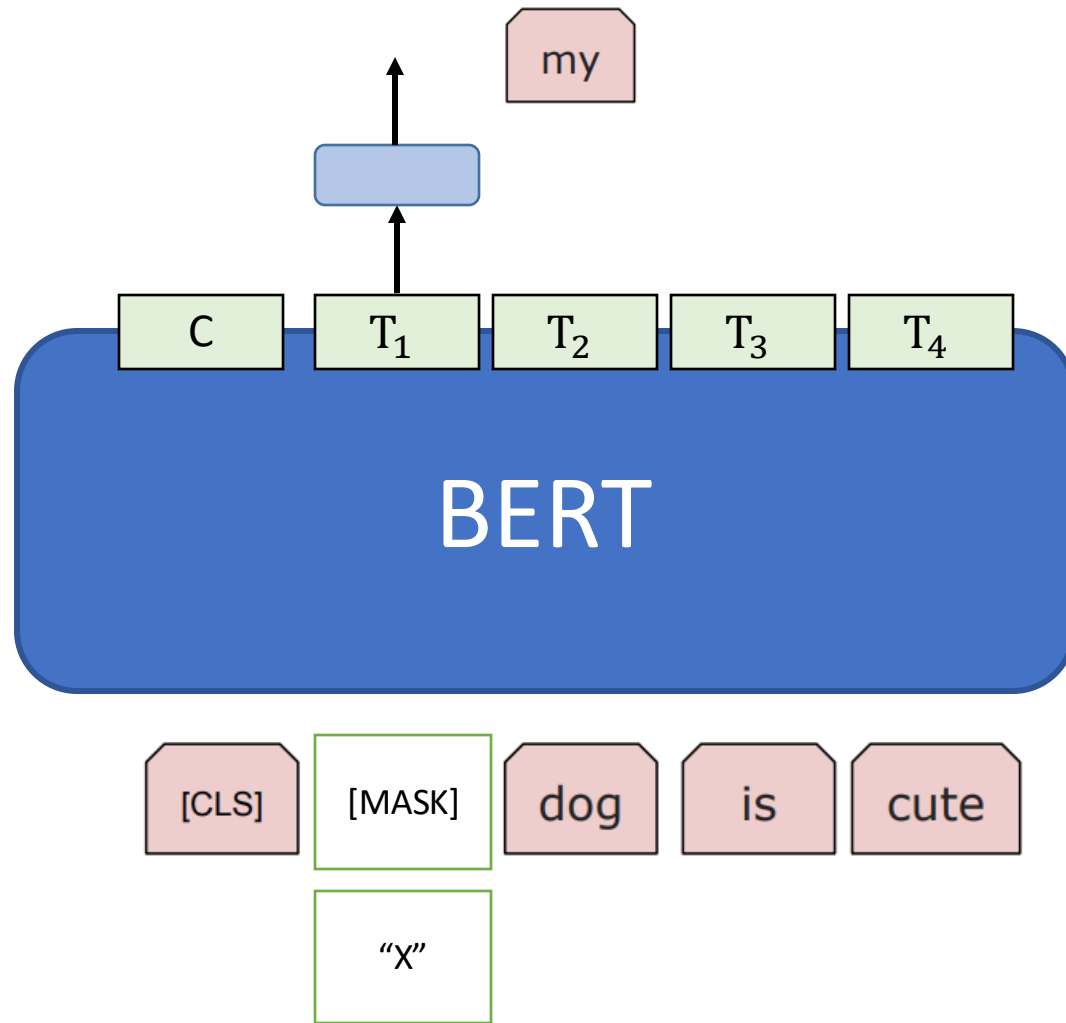
Illegally injecting future label information to LM.

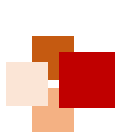
Masked language model task



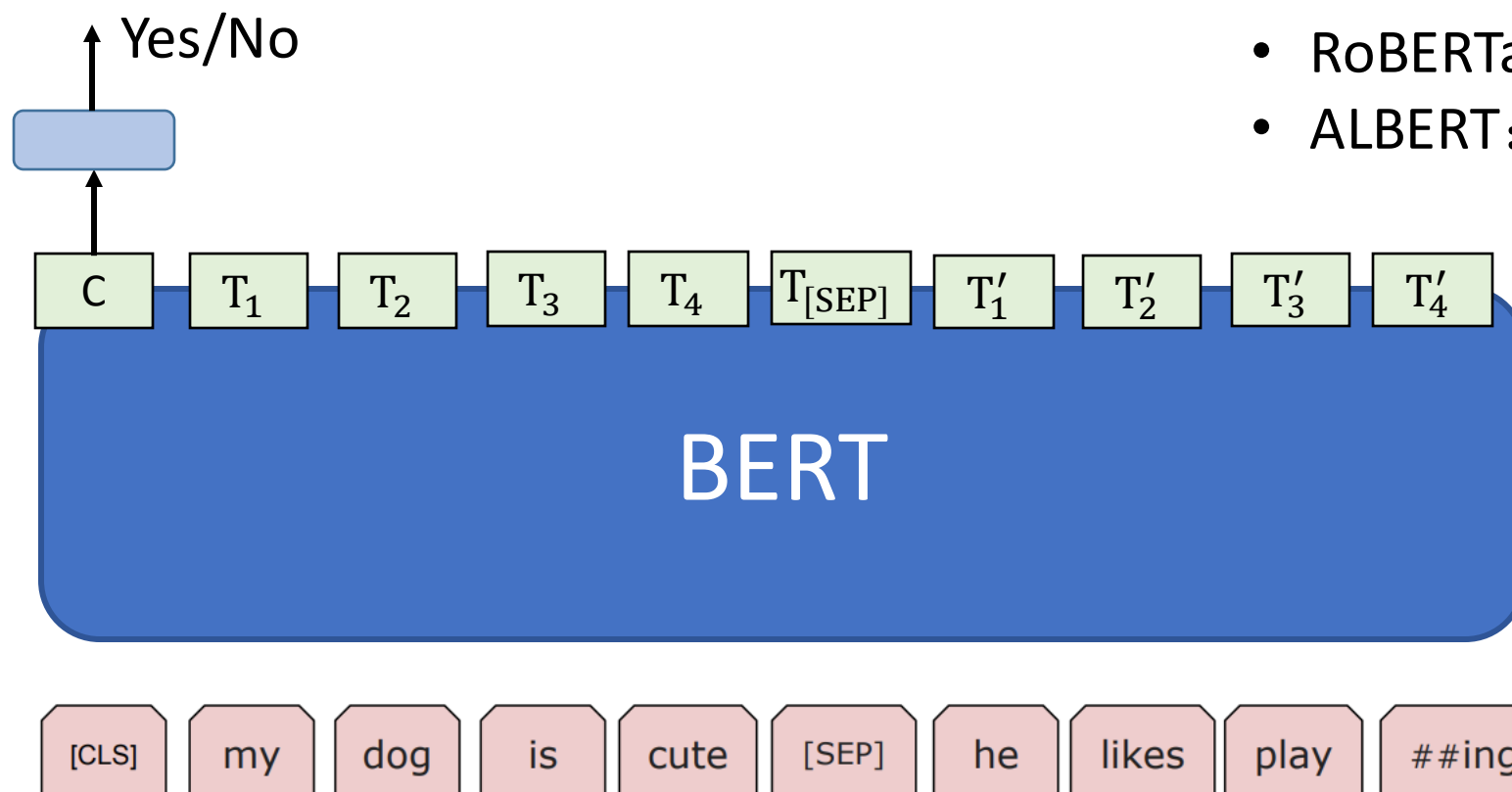
Finetuning

Pre-training: Masked LM





Pre-training: Next Sentence Prediction



- RoBERTa: not helpful
- ALBERT: Sentence order prediction

Liu, Yinhan, et al. "Roberta: A robustly optimized bert pretraining approach." 2019

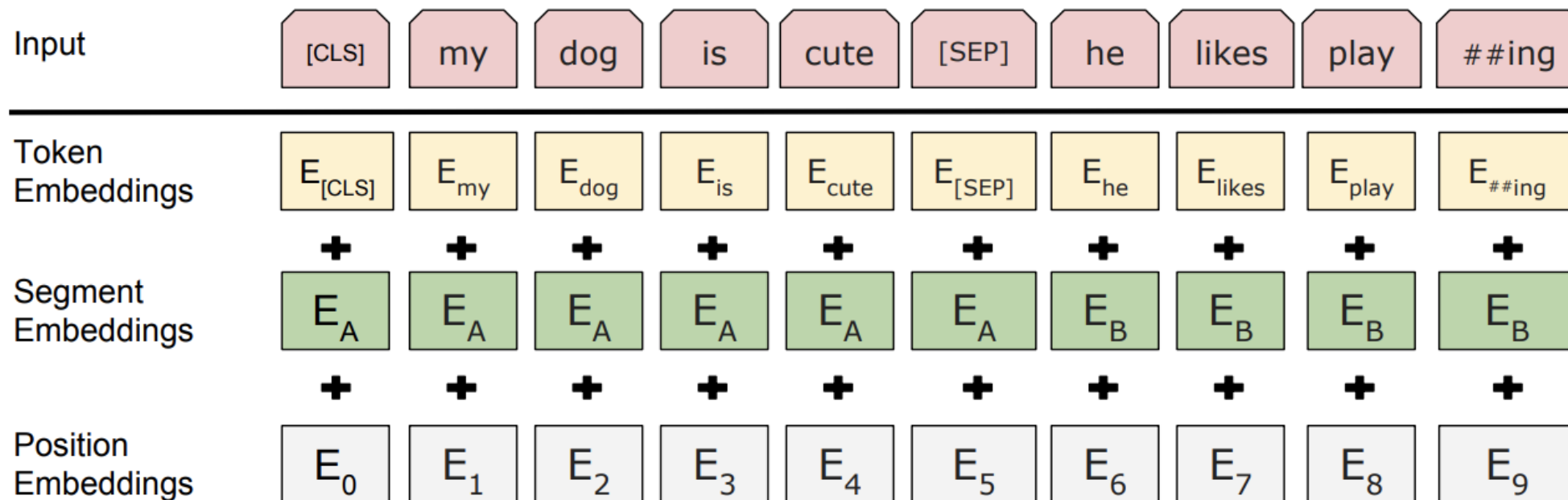
Lan, Zhenzhong, et al. "Albert: A lite bert for self-supervised learning of language representations." 2019



Embeddings



30000 WordPiece





Model Details



- Data: Wikipedia (2.5B words) + BookCorpus (800M words)
- Batch Size: 131,072 words (1024 sequences * 128 length or 256 sequences * 512 length)
- Training Time: 1M steps (~40 epochs)
- Optimizer: AdamW, $1e-4$ learning rate, linear decay
- BERT-Base: 12-layer, 768-hidden, 12-head
- BERT-Large: 24-layer, 1024-hidden, 16-head
- Trained on 4x4 or 8x8 TPU slice for 4 days



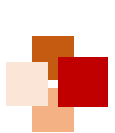
- CoLA: The Corpus of Linguistic Acceptability
- SST-2: The Stanford Sentiment Treebank
- MRPC: Microsoft Research Paraphrase Corpus
- STS-B: Semantic Textual Similarity Benchmark
- QQP: Quora Question Pairs
- MNLI: Multi-Genre Natural Language Inference
- QNLI: Question Natural Language Inference
- RTE: Recognizing Textual Entailment
- WNLI: Winograd Natural Language Inference



GLUE Results

System	MNLI-(m/mm) 392k	QQP 363k	QNLI 108k	SST-2 67k	CoLA 8.5k	STS-B 5.7k	MRPC 3.5k	RTE 2.5k	Average
Pre-OpenAI SOTA	80.6/80.1	66.1	82.3	93.2	35.0	81.0	86.0	61.7	74.0
BiLSTM+ELMo+Attn	76.4/76.1	64.8	79.9	90.4	36.0	73.3	84.9	56.8	71.0
OpenAI GPT	82.1/81.4	70.3	88.1	91.3	45.4	80.0	82.3	56.0	75.2
BERT _{BASE}	84.6/83.4	71.2	90.1	93.5	52.1	85.8	88.9	66.4	79.6
BERT _{LARGE}	86.7/85.9	72.1	91.1	94.9	60.5	86.5	89.3	70.1	81.9

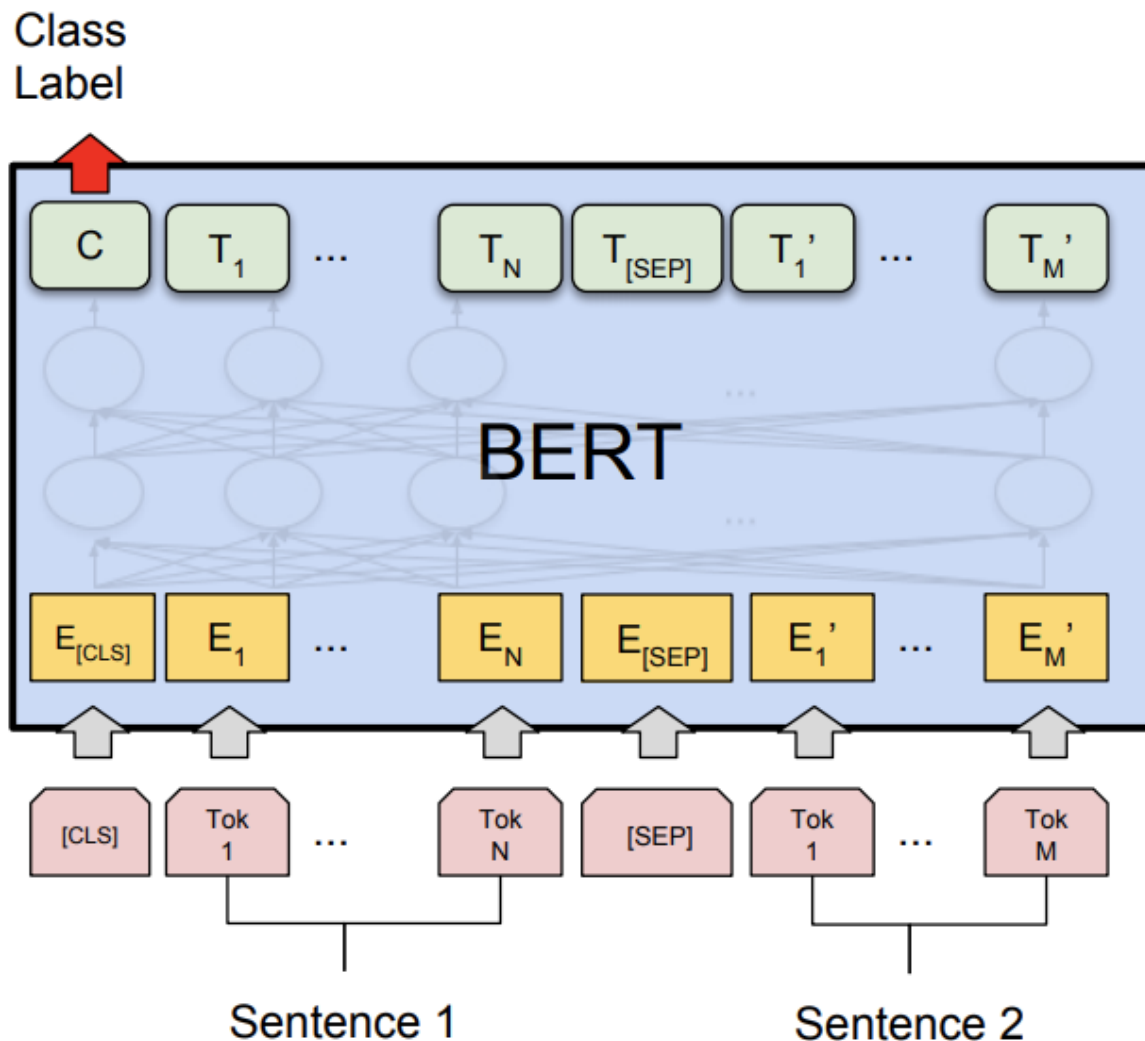


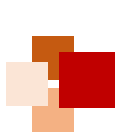


Fine-tuning: Sentence Pair Classification Tasks

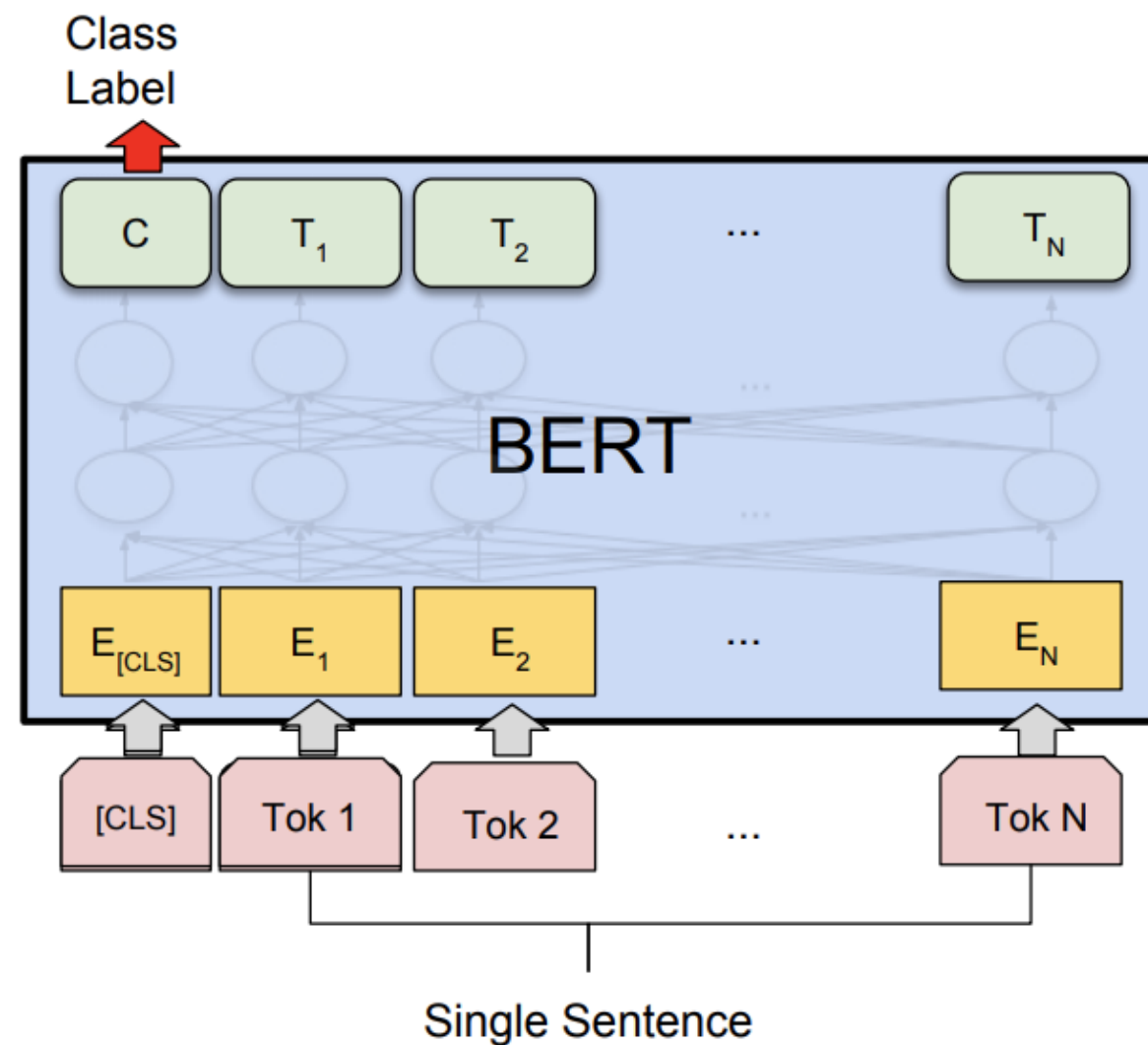


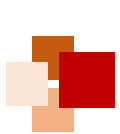
random
initialization



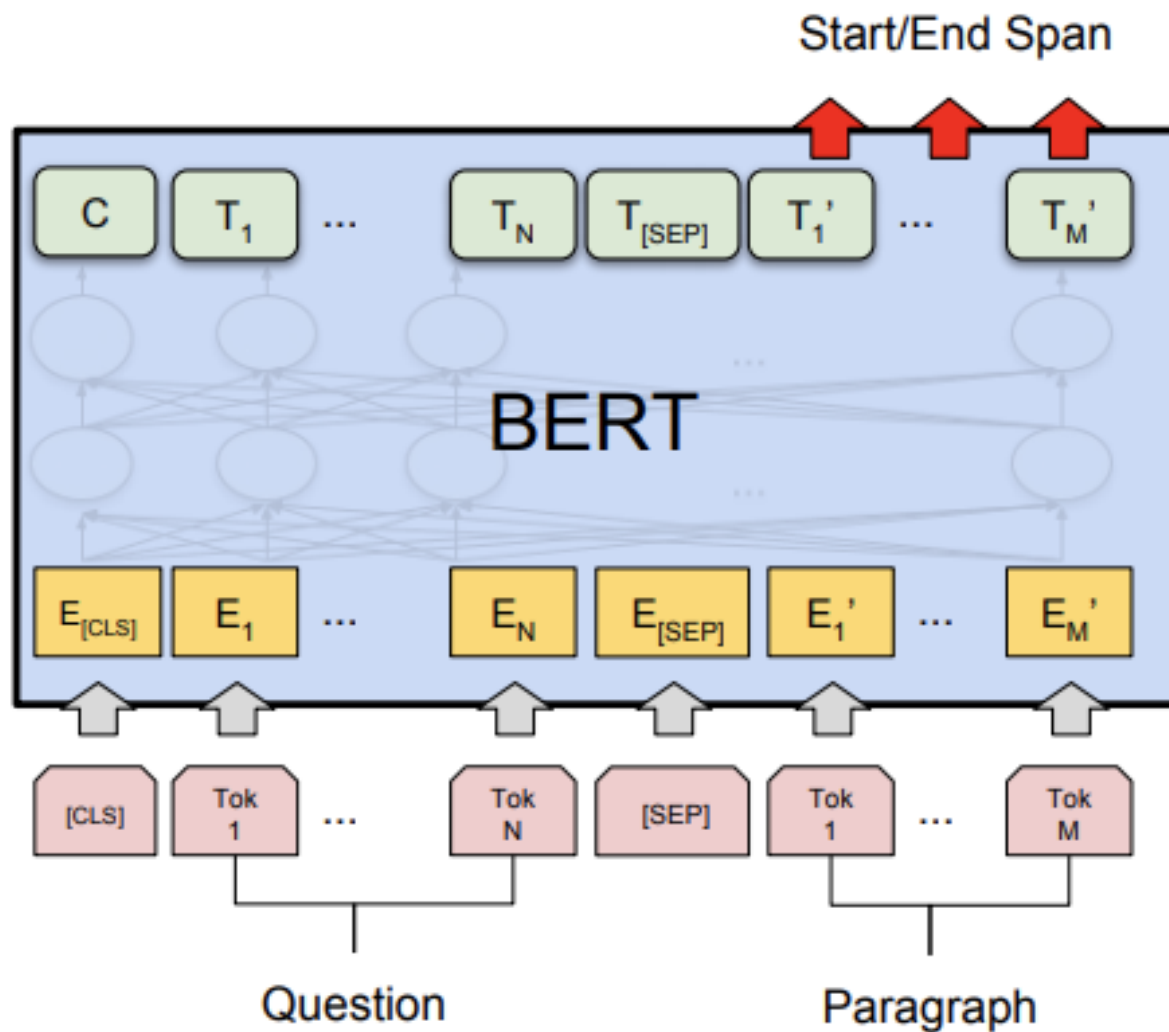


Fine-tuning: Single Sentence Classification





Fine-tuning: Question Answering Tasks

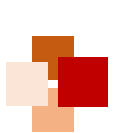


Random Initialized vectors:

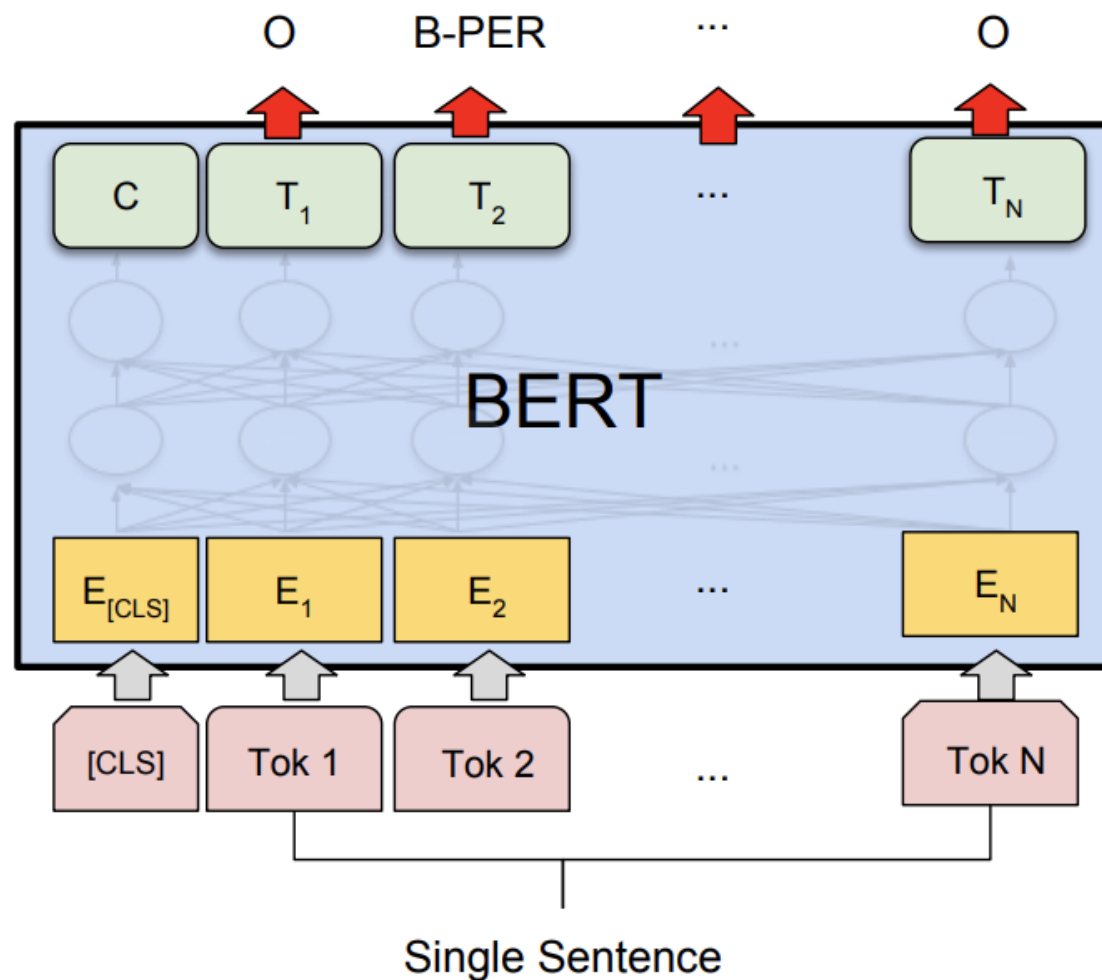
$$v_s, v_e$$

$$p_i = \frac{\exp(v_s^T T'_i)}{\sum \exp(v_s^T T'_i)}$$





Fine-tuning: Single Sentence Tagging Tasks



- More epochs, more data, bigger models
- Masking
- Relative position embeddings
- Factorized embedding parameterization
- Cross-layer parameter sharing
- *Many more ...*



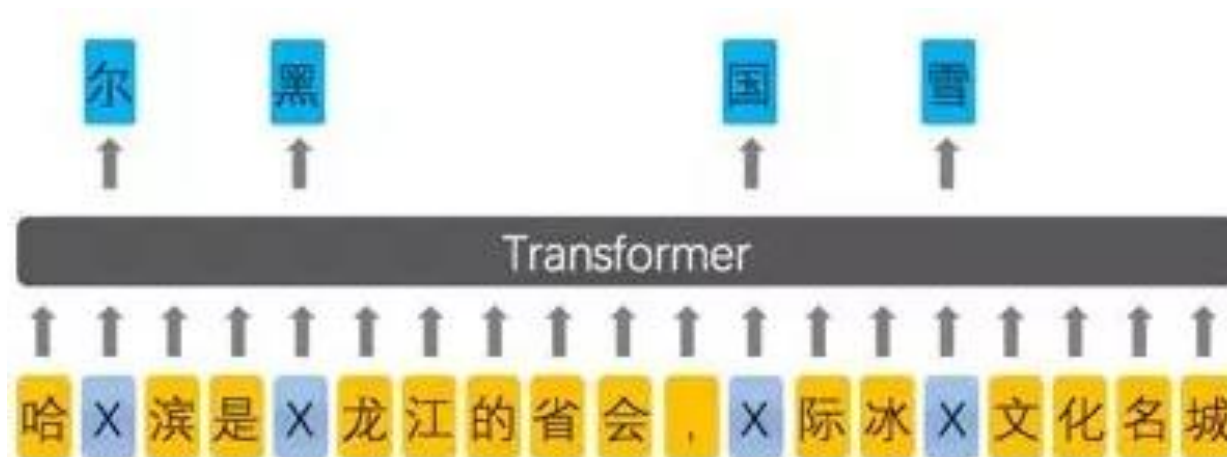
Enhanced Representation through Knowledge Integration (ERNIE)



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BERT

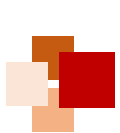


ERNIE



Source of image: <https://zhuanlan.zhihu.com/p/59436589>

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Multilingual BERT



Trained on 104 languages



Task specific training
data for English

Task specific testing
data for Chinese

BERT	英语	中文	西班牙语	德语	阿拉伯语	乌尔都语
Translate-Train	81.4	74.2	77.3	75.2	70.5	61.7
Translate-Test	81.4	70.1	74.9	74.4	70.4	62.1
mBERT	81.4	63.8	74.3	70.5	62.1	58.3



Class 3



?



?

The Surprising Cross-Lingual Effectiveness of BERT

A Study of Cross-Lingual Ability and Language-specific Information in Multilingual BERT